

First Version of the Multiplication Algorithm and Hardware

Iteration	Step	Multiplier	Multiplicand	Product
0	Initial values	0011	0000 0010	0000 0000
1	1a: $1 \Rightarrow \text{Prod} = \text{Prod} + \text{Mcand}$	0011	0000 0010	0000 0010
	2: Shift left Multiplicand	0011	0000 0100	0000 0010
	3: Shift right Multiplier	0001	0000 0100	0000 0010
2	1a: $1 \Rightarrow \text{Prod} = \text{Prod} + \text{Mcand}$	0001	0000 0100	0000 0110
	2: Shift left Multiplicand	0001	0000 1000	0000 0110
	3: Shift right Multiplier	0000	0000 1000	0000 0110
3	1: $0 \Rightarrow$ no operation	0000	0000 1000	0000 0110
	2: Shift left Multiplicand	0000	0001 0000	0000 0110
	3: Shift right Multiplier	0000	0001 0000	0000 0110
4	1: $0 \Rightarrow$ no operation	0000	0001 0000	0000 0110
	2: Shift left Multiplicand	0000	0010 0000	0000 0110
	3: Shift right Multiplier	0000	0010 0000	0000 0110

Iteration	Step	Multiplier	Multiplicand	Product
0	Initial values	0011	0000 0010	0000 0000
1	1a: $1 \Rightarrow \text{Prod} = \text{Prod} + \text{Mcand}$	0011	0000 0010	0000 0010
	2: Shift left Multiplicand	0011	0000 0100	0000 0010
	3: Shift right Multiplier	0001	0000 0100	0000 0010
2	1a: $1 \Rightarrow \text{Prod} = \text{Prod} + \text{Mcand}$	0001	0000 0100	0000 0110
	2: Shift left Multiplicand	0001	0000 1000	0000 0110
	3: Shift right Multiplier	0000	0000 1000	0000 0110
3	1: $0 \Rightarrow$ no operation	0000	0000 1000	0000 0110
	2: Shift left Multiplicand	0000	0001 0000	0000 0110
	3: Shift right Multiplier	0000	0001 0000	0000 0110
4	1: $0 \Rightarrow$ no operation	0000	0001 0000	0000 0110
	2: Shift left Multiplicand	0000	0010 0000	0000 0110
	3: Shift right Multiplier	0000	0010 0000	0000 0110

Using 4-bit numbers to save space, multiply $2_{\text{ten}} \times 3_{\text{ten}}$, or $0010_{\text{two}} \times 0011_{\text{two}}$.

Work will continue till not found 0000.

Final Version of the Multiplication Algorithm and Hardware:

Iteration	Step	Multiplicand	Product
0	Initial values	0010	0000 0011
1	1a: 1 => Prod = Prod + Mcand	0010	0010 0011
	2: Shift right Product	0010	0001 0001
2	1a: 1 => Prod = Prod + Mcand	0010	0011 0001
	2: Shift right Product	0010	0001 1000
3	1: 0 => no operation	0010	0001 1000
	2: Shift right Product	0010	0000 1100
4	1: 0 => no operation	0010	0000 1100
	2: Shift right Product	0010	0000 0110

Multiply example using third algorithm in Figure 4.32.

If the product's last digit is 1, then $\text{Prod} = \text{Prod} + \text{Mcand}$